



CAdam

Context-Adaptive Moment Estimation for 3D Gaussian Densification in Generative Distillation



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Verify the Signal → Control the Growth → 97% Fewer Gaussians

Gaussians Reduction

↓ **97.46%**



"A chubby Shiba Inu puppy with round eyes, soft fur texture, Disney animated movie style, happy expression, 3D render, bright lighting."

Gaussians Reduction

↓ **96.46%**



"A stylized chunky toy biplane, cheerful red and cream colors, soft plastic texture, cartoon proportions, whimsical game vehicle."

Gaussians Reduction

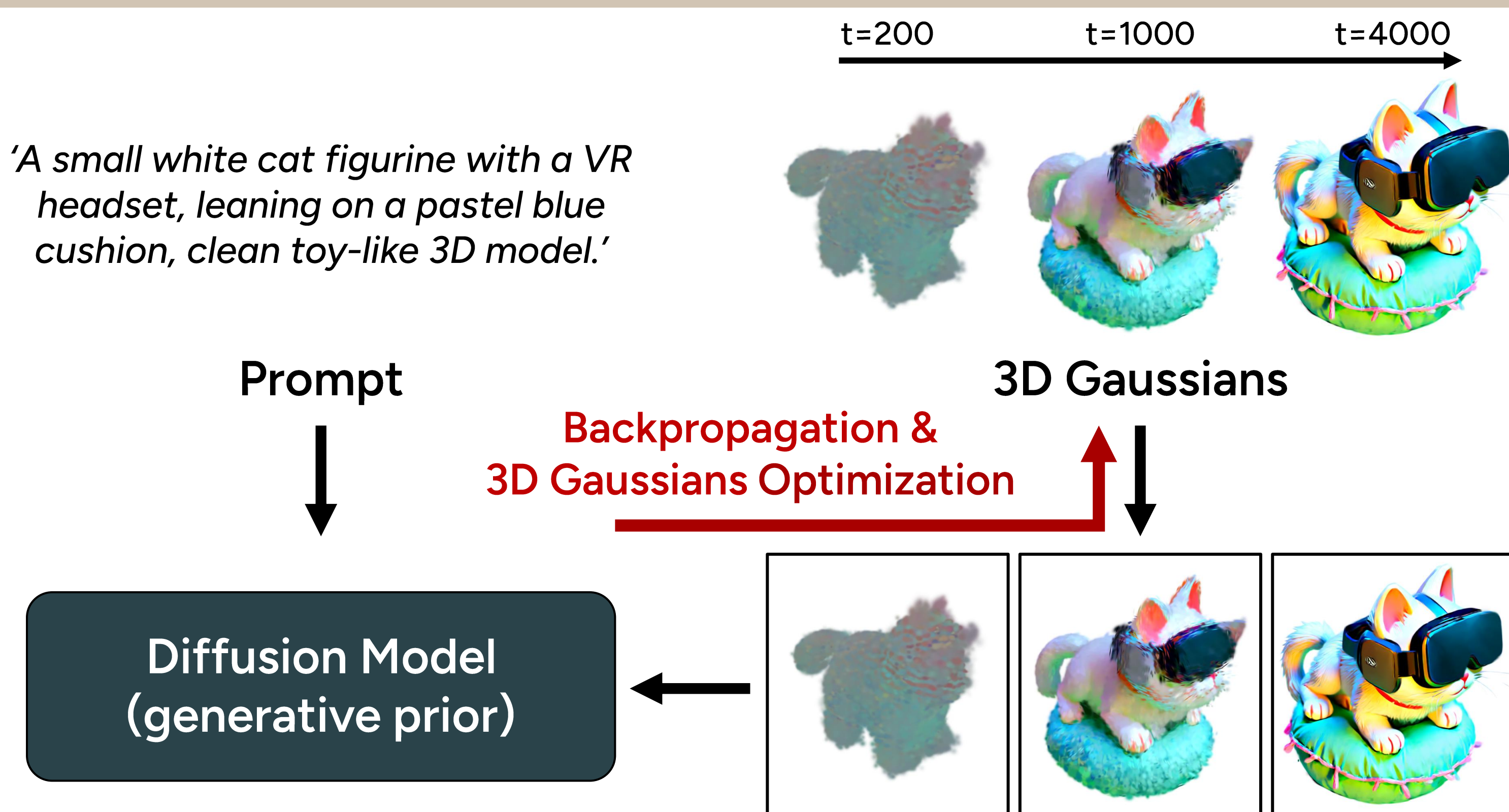
↓ **97.41%**



"a steampunk owl with mechanical wings, intricate gears, brass, steampunk, realistic, HDR, photorealistic, 8K."

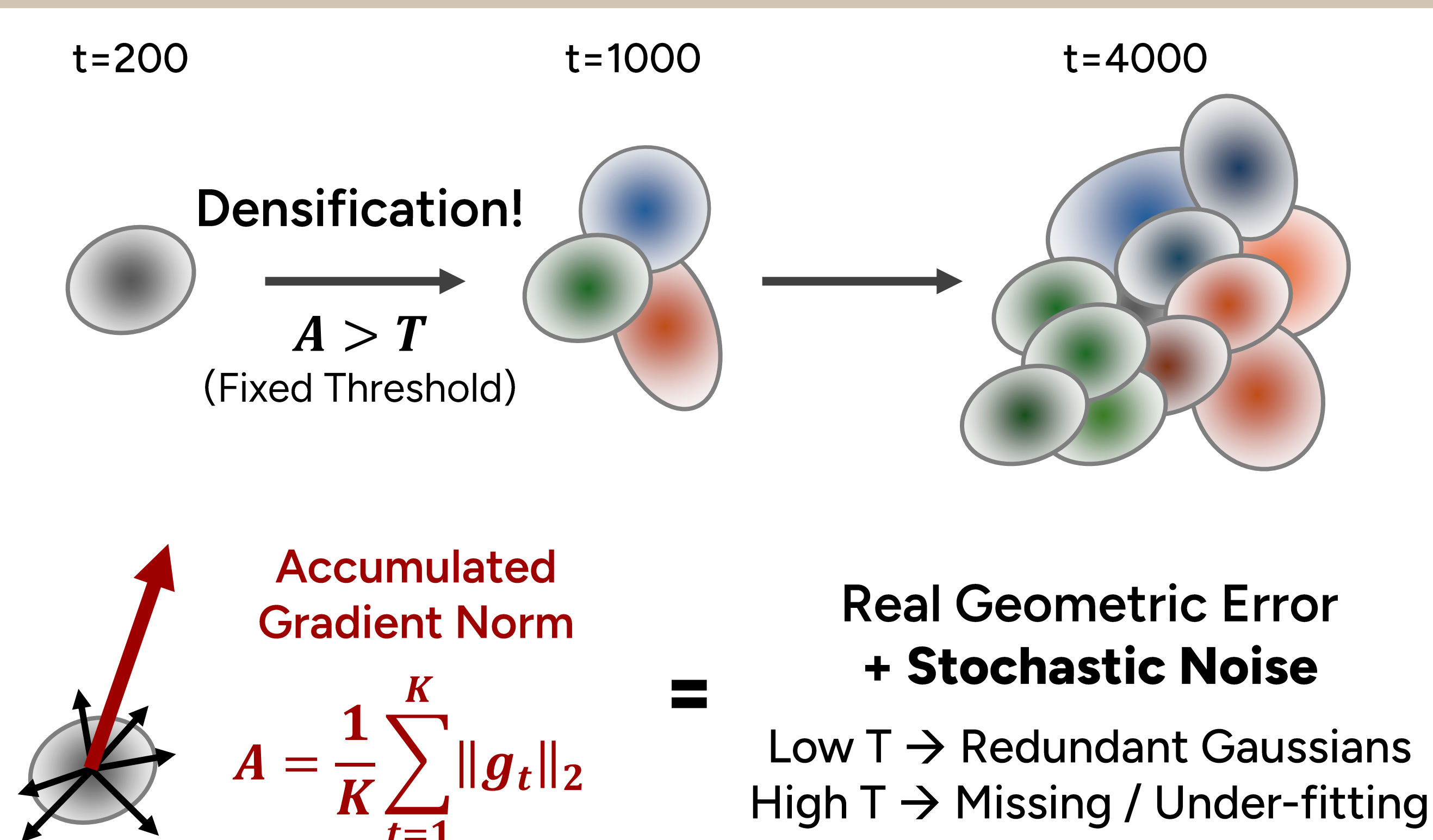
Background

Optimization-Based Generative 3DGS Pipeline



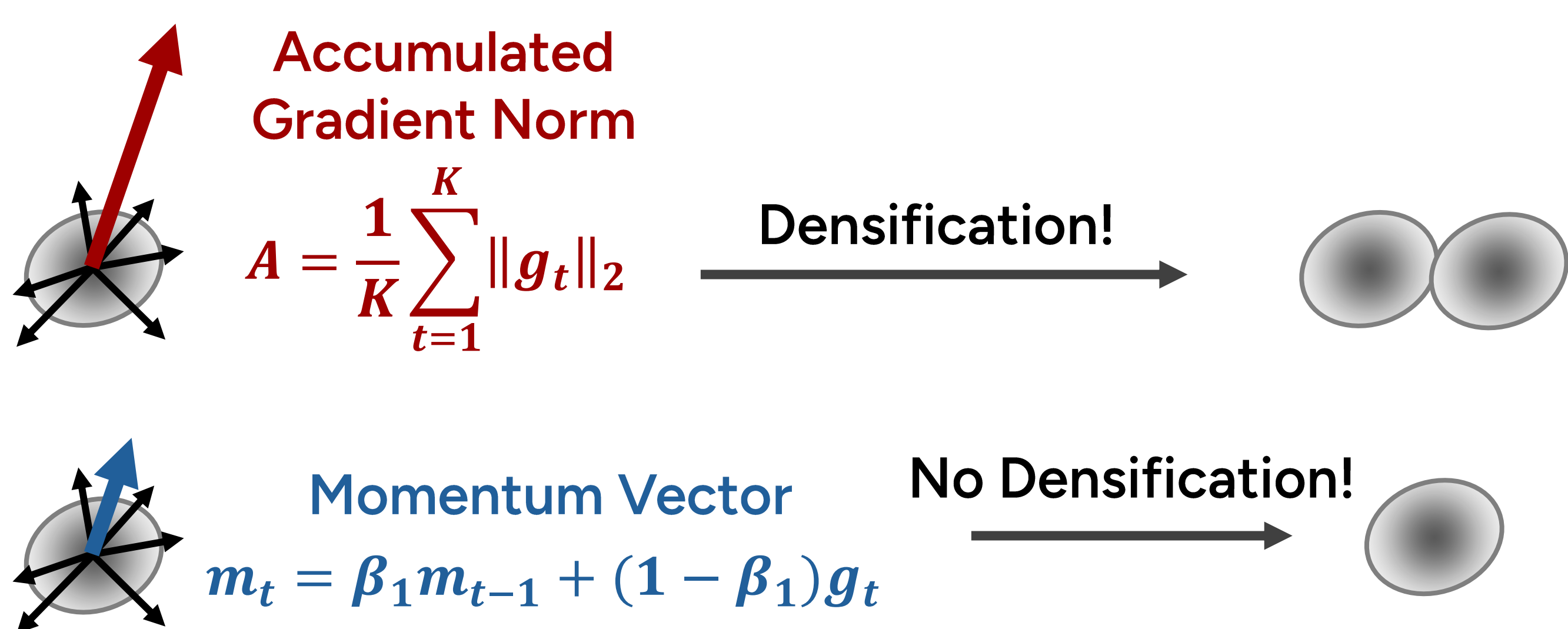
Problem

Densification Dilemma



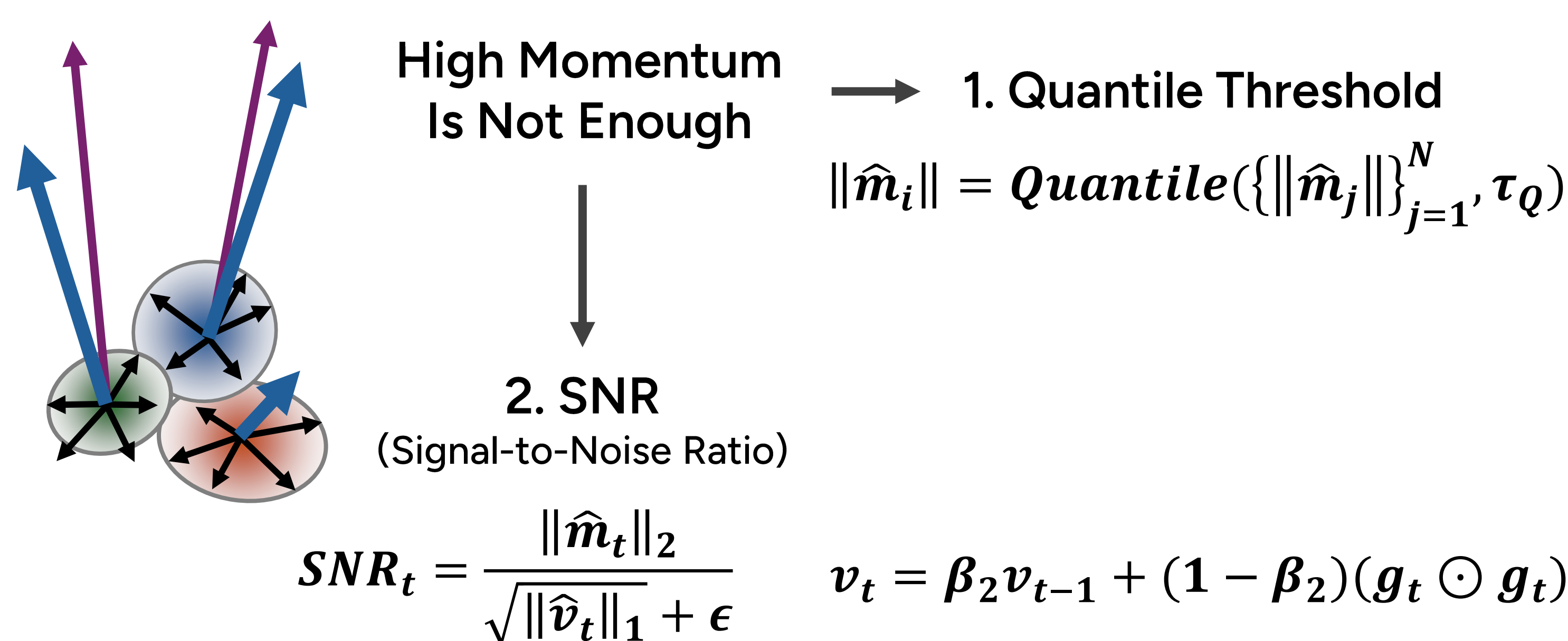
Our Method

Momentum as Signal Verification



Our Method

Context-Adaptive Selection



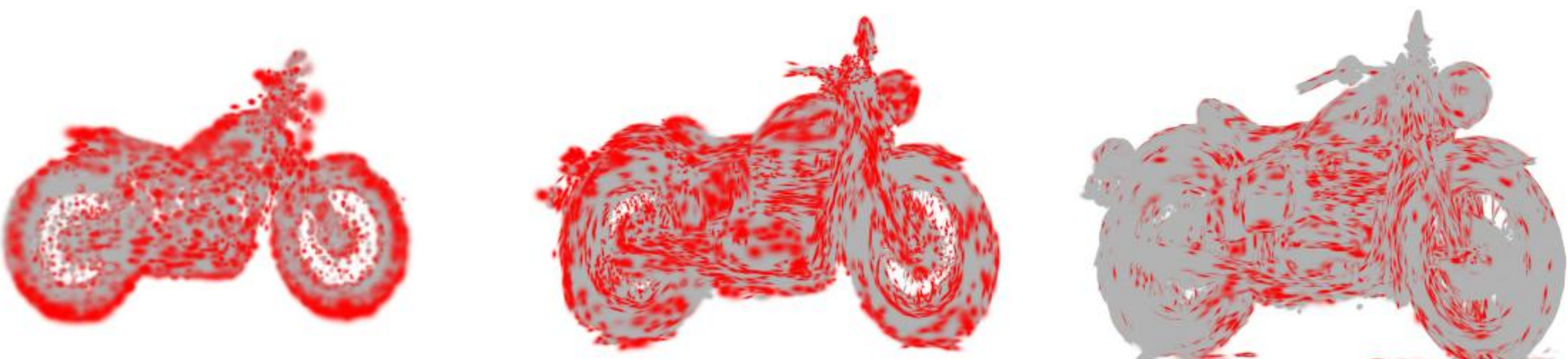
Results

Optimization Dynamics

Densification Candidate Masking (Baseline)



Densification Candidate Masking (CAdam)



Results

Threshold Variation

Balanced-Densification ($\tau_Q = 0.9 \mid \tau_{SNR} = 0.1$)



Over-Densification ($\tau_Q = 0.3 \mid \tau_{SNR} = 0.05$)

